NITROSEND — PRODUCT ENGINEER APPLICATION

The missing half of the send API

Nitrosend tells you a message was accepted. It never tells you what happened next. This is a working prototype of the piece that's missing: outbound webhooks for suppression events.

AUTHOR
Jizhe Hou

STAGING
nitrosend-webhooks-demo.fly.dev

REPOSITORY
github.com/partwith/nitrosend-webhooks-demo

STACK
Rails 8.1 · Postgres · Redis · GoodJob · Vue 3 · Tailwind

THE GAP

A bounce happens, and only Nitrosend finds out

When a send bounces or draws a spam complaint, Nitrosend does the right thing internally: SES reports the event, the address goes on the suppression list, and the account stops sending to it. That protects Nitrosend's own domain reputation.

But the event stops there. Nothing tells the application that triggered the send.

Take a concrete case: a booking platform calls the transactional API to send a confirmation. Nitrosend accepts the message, queues it, returns a message ID. From the caller's perspective the send succeeded — that's the last thing its own database knows. If the address hard-bounces an hour later, the caller has no way of finding out unless someone manually checks the Nitrosend dashboard. The customer who never got their confirmation, and the support ticket that follows, both trace back to this gap.

This isn't a hypothetical integration problem — it's the exact scenario the product is built around. Nitrosend's own positioning is *"connect once, and your AI assistant can manage your entire email operation"* for downstream apps and agents. Suppression state is exactly the kind of operational fact a downstream system needs back, and right now it doesn't get it.

WHAT'S ADVERTISED VS. WHAT'S THERE

The marketing copy describes a feature that doesn't exist yet

Worth being precise about this before treating it as a finding — checked against both the marketing site and the live API surface, not assumed.

CLAIM	WHERE	REALITY
"Real-time event webhooks for delivery, opens, clicks, bounces, and more"	`/features`	✗ No webhook resource in the API
"Sent, delivered, opened, clicked, bounced, complained, unsubscribed, failed… signed with HMAC-SHA256, retried with exponential backoff"	`/transactional-email-api`	✗ Described in detail, not present
Suppression list exists and is enforced	Product behaviour	✓ Confirmed — sends to suppressed addresses are blocked
Suppression list is queryable	`GET /v1/my/suppressions`	✓ Confirmed — read-only, no way to subscribe to changes
Webhook configuration surface (register endpoint, choose events, view delivery attempts)	Dashboard + full API reference	✗ Not present anywhere

The full API reference lists every other configuration surface — Integrations, Domains, Suppressions — and none of them is a webhook. This reads less like a roadmap item and more like documentation that shipped ahead of the feature.

SCOPE

What this prototype covers

Narrowly: outbound **suppression events** — bounce and complaint notifications pushed to a downstream endpoint — rather than the full `sent/delivered/opened/clicked` taxonomy the marketing copy implies.

A bounce or complaint is actionable in a way opens and clicks aren't — retry with a different address, flag a contact record, alert a human. It's also a self-contained, demoable slice that maps directly onto data Nitrosend already has, rather than requiring new tracking infrastructure. Per-brand endpoint scoping beyond a single account and UI for replaying historical events are deliberately out of scope for a prototype.

WHAT I BUILT

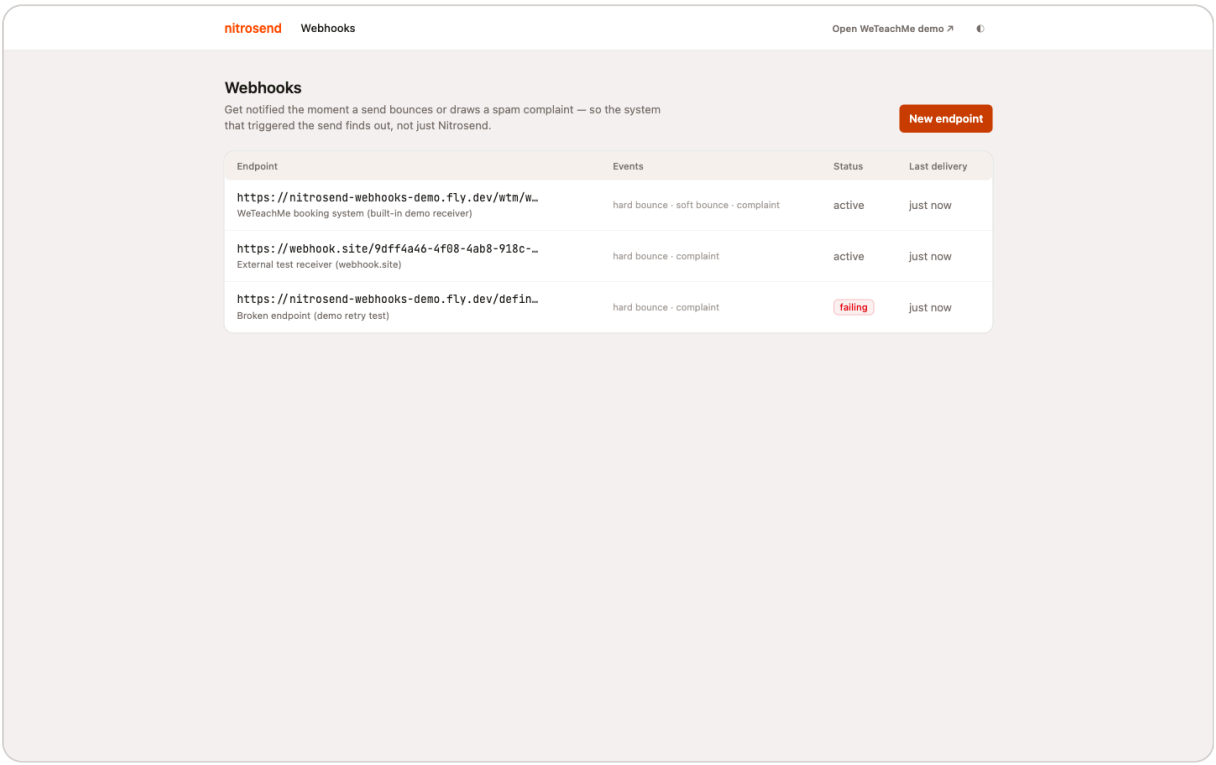
The whole loop, working end to end

Rather than mock the feature, I built the real pipeline: a small "WeTeachMe" booking app stands in for any downstream integrator, sending confirmations through a Nitrosend-shaped API. When one bounces, a real signed HTTP request goes out, gets verified, and the booking updates live — no refresh.

1

Register an endpoint

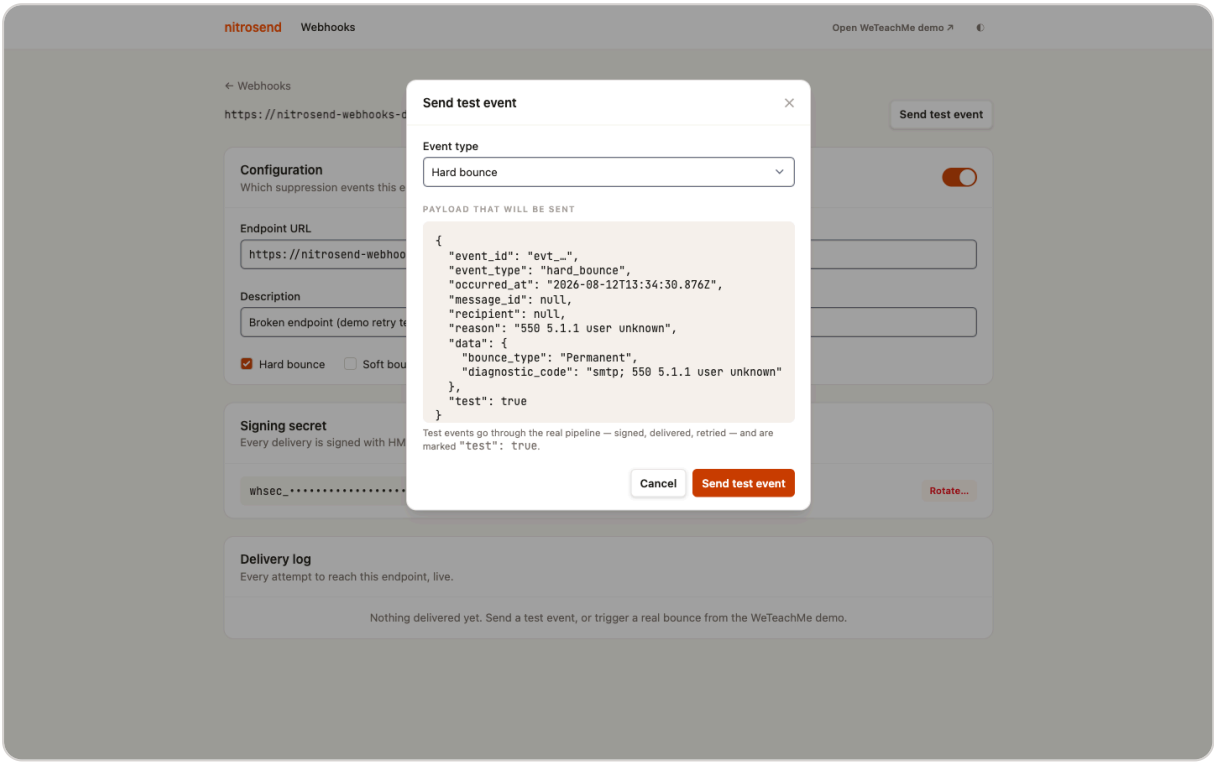
A URL, which suppression events it should receive, an enable toggle. The events column stays plain text — it's passive configuration, not something that needs attention, so it doesn't get a badge.



Three endpoints, one list. The seeded receiver, an external `webhook.site` URL for inspecting raw payloads, and one pointed at a dead path — its `failing` badge is the only colour on the page that isn't the primary action, because it's the only row that needs attention.

2 Preview before you send

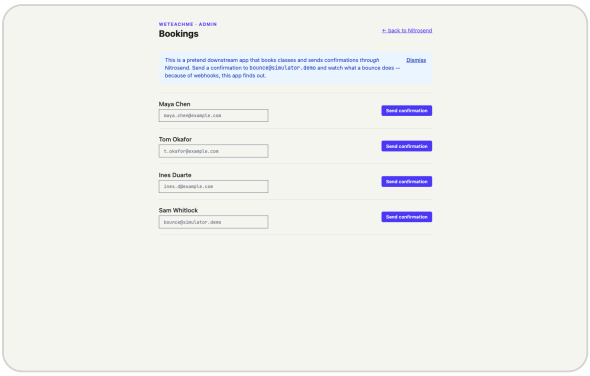
A test event goes through the real pipeline — signed, delivered, retried on failure — and is marked `"test": true` so it's distinguishable from a real bounce, but otherwise indistinguishable in behaviour.



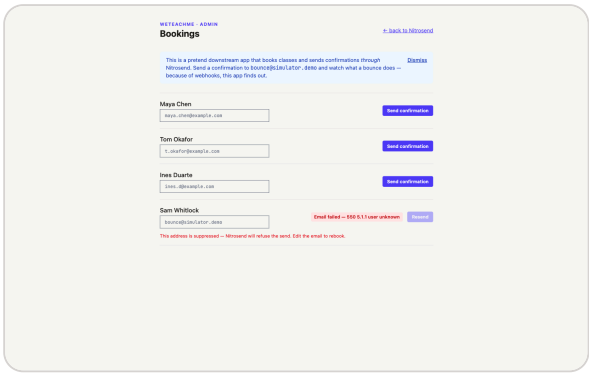
No surprises at send time. The exact payload is visible before you commit to firing it.

3 Trigger a real bounce

The demo simulates SES with magic addresses — `bounce@simulator.demo` hard-bounces, `complaint@simulator.demo` files a spam complaint — so the whole loop is reproducible without touching a real mailbox.



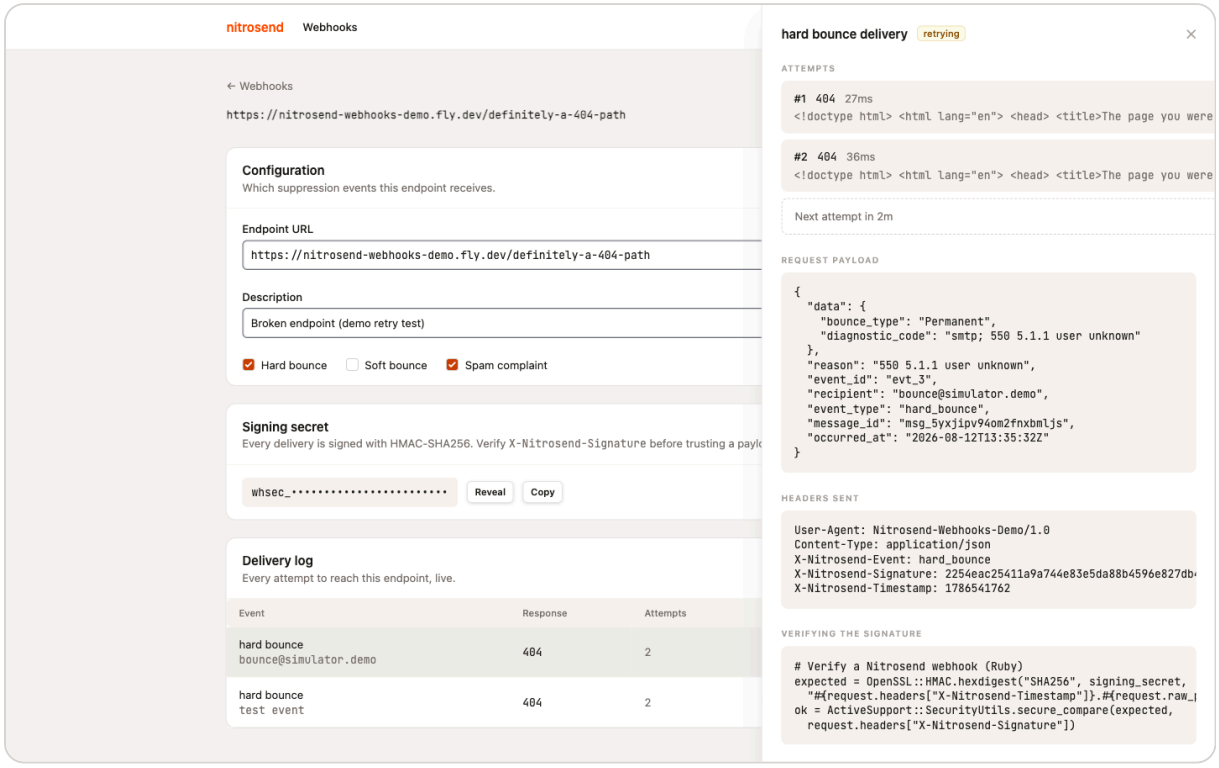
Before — Sam Whitlock's confirmation hasn't been sent yet.



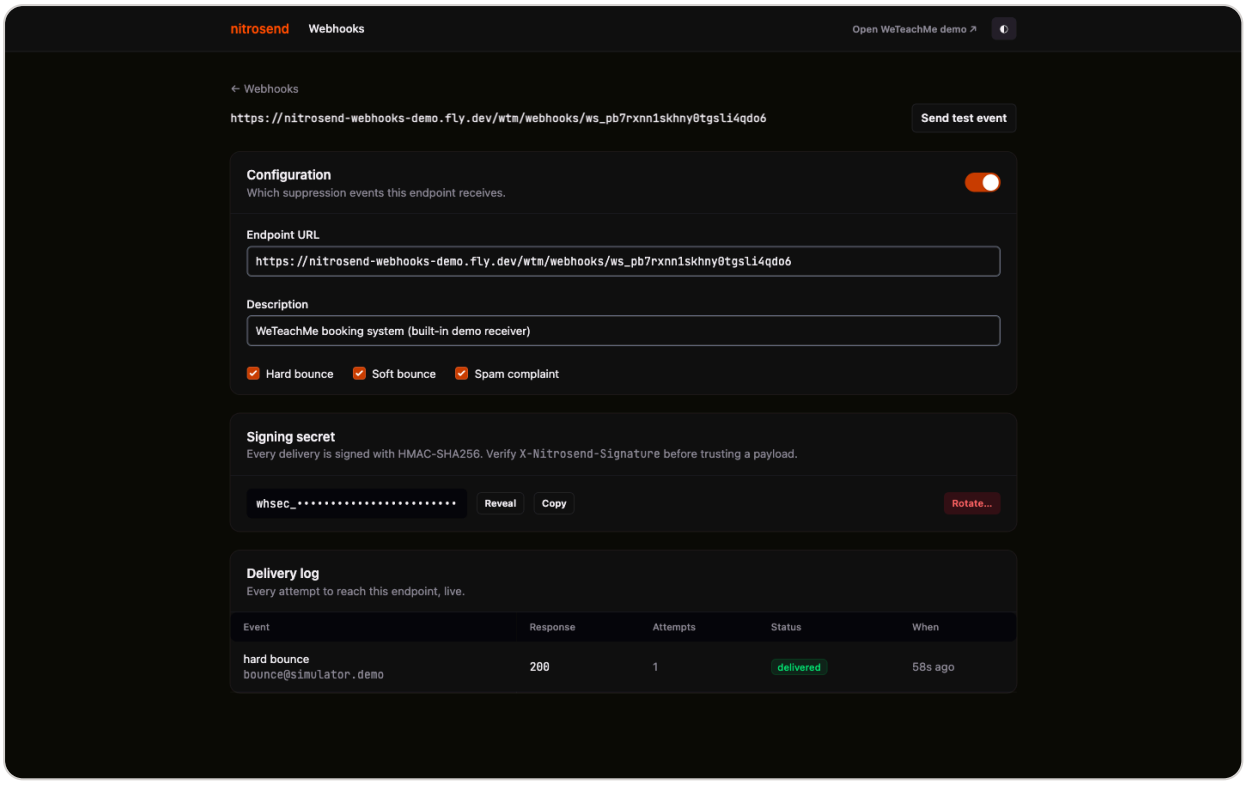
~5 seconds later, unprompted — flagged, with the reason, and the address is now blocked from resending.

4 See the delivery, not just the result

Every attempt is logged with its real response — status code, latency, body — so a retry in progress is visible as it happens, not summarised after the fact.



This is the deliberately-broken endpoint, mid-retry. Two real attempts, two real 404 bodies, a live countdown to the third. The panel also shows the exact headers sent and the code to verify them — because "here's a signature" is only useful if verifying it is a five-minute job, not a support ticket.



Dark mode isn't an afterthought — same information density, same restraint.

ENGINEERING NOTES

The parts that don't show up in a screenshot

Signing, for real

Every delivery carries `X-Nitrosend-Event`, `X-Nitrosend-Timestamp`, and `X-Nitrosend-Signature`. The signature is `HMAC-SHA256(secret, "{timestamp}.{body}")`, checked with a constant-time comparison and a ±5 minute timestamp window to block replay. Rotating a secret doesn't break in-flight deliveries — the previous secret keeps verifying for 24 hours.

Retries that actually happen

Failed deliveries retry at 30s, 2m, 10m, 30m — five attempts, then a terminal `failed` state. Not simulated: the delivery log above is reading real `DeliveryAttempt` rows written by a real job that really waited.

A public demo that makes real HTTP calls

Since anyone can register an endpoint URL, every delivery is pinned to a DNS-resolved IP that's checked against private/reserved ranges before the request goes out — closing the gap between "this URL looked safe" and "this URL is what we actually connected to."

Every visitor gets their own sandbox

No login. A signed cookie assigns a fresh, isolated workspace on first visit, seeded with demo data, purged after 24 hours of inactivity. Two reviewers can use the staging link at once without seeing each other's clicks.

```ruby
# verifying a delivery (Ruby, but the math is the point)
expected = OpenSSL::HMAC.hexdigest("SHA256", signing_secret,
  "#{request.headers['X-Nitrosend-Timestamp']}.#{request.raw_post}")
ok = ActiveSupport::SecurityUtils.secure_compare(expected,
  request.headers['X-Nitrosend-Signature'])
```

TRY IT

60 seconds, start to finish

Open the staging link, go to **WeTeachMe**, send a confirmation to one of the addresses below, then flip back to **Nitrosend** and watch the delivery log. Every visitor gets a clean, private sandbox — nothing you do affects anyone else looking at the same link.

ADDRESS	WHAT HAPPENS
`bounce@simulator.demo`	Hard bounce → suppressed → `hard_bounce` webhook fires
`softbounce@simulator.demo`	Soft bounce → not suppressed → `soft_bounce` webhook fires
`complaint@simulator.demo`	Spam complaint → suppressed → `complaint` webhook fires
anything else	Delivers normally, no event

Jizhe Hou — Nitrosend Product Engineer application

Built with Rails 8.1, Vue 3, and the product's own design system